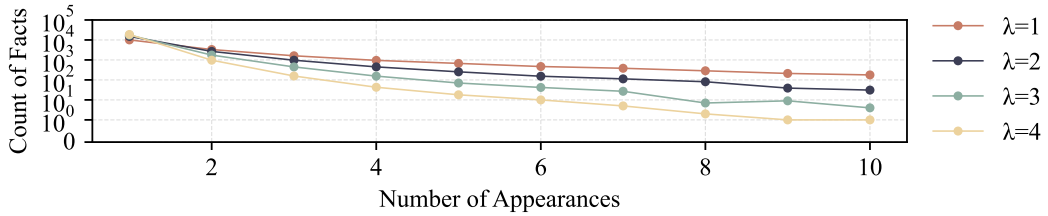

Fig. 1. Illustration of fact repetition frequencies across varying parameters (γ). Plot shows how often a fact appears (x-axis) versus how many facts have that appearance count (y-axis, logarithmic scale).

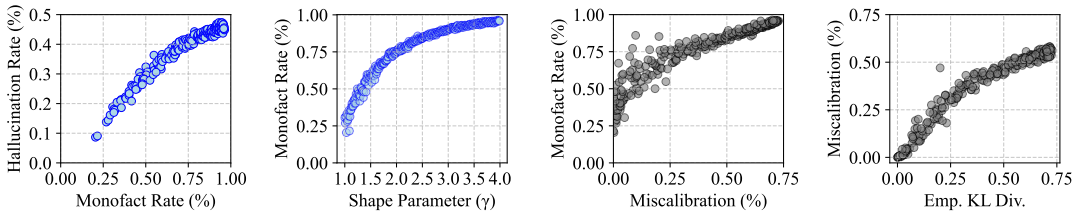

Fig. 2. Each dot represents a sample of statements. **Left:** Results show a positive relationship between monofact rate and hallucination. **Middle Left:** Heavy-tailed document distributions yield lower monofact rates. **Middle Right:** Miscalibration increases with monofact rate, suggesting better learning and calibration in low-monofact distributions. **Right:** Strong positive correlation between empirical KL divergence and miscalibration metrics.

A. Statement Generation Process. Let g denote the distribution over six-tuples $x = (t_1, \ldots, t_6)$. For any pair of tokens (t_1, t_2), the bigram model learns $g(t_1, t_2) = g(t_1)\, g(t_2 \mid t_1)$, where $g(t_1)$ is the marginal over first tokens and $g(t_2 \mid t_1)$ the conditional over second tokens. Generating a new six-tuple proceeds sequentially, giving $g(x) = g(t_1) \prod_{i=1}^{5} g(t_{i+1} \mid t_i)$.

To be more specific, we first draw $t_1 \sim g(t_1)$ and then, for each position $i = 1{:}5$, sample $t_{i+1} \sim g\big(t_{i+1} \mid t_i\big)$. Note that while this approach is similar to standard bigram language modeling, it does not exploit the full structure of our movie data where each position has its own distinct token space. Instead, we treat all position pairs equivalently in building our transition probabilities. For evaluation, each model is trained on $|S| = 10,000$ statements and generates an equal number of new statements.

B. Controlled Miscalibration Injection. To dissect the role of miscalibration while *holding the monofact rate fixed*, we introduce a lightweight *upweighting* routine that perturbs the learned transition counts of a trained bigram model to inject miscalibration into g.

Let the initial bigram counts be $C_0(t_i, t_j)$. Choose a subset $E_k \subset S$ of size k. For every token pair (t_i, t_j) occurring in E_k, add one extra count: $C_1(t_i, t_j) \leftarrow C_0(t_i, t_j) + \mathbf{1}_{(t_i, t_j) \in E_k}$. The same increment is applied to initial-state counts. Fig. S1 in the SI Appendix details this update with comprehensive steps and then calls the normalization subroutine to convert counts back into normalized probabilities.

Upweighting only a fraction of the training data induces selective over-confidence, while keeping the empirical monofact rate constant. As k grows, miscalibration and hallucination first increase and decline to baseline once $E_k = S$ (the global duplication cancels after normalization). This controlled knob lets us chart hallucination as a function of miscalibration.

3. n-Gram Model Results

A. Monofact and Hallucination. Our controlled simulations validate the monofact-hallucination relationship in Eq. (1). The left panel of Fig. 2 shows an approximately linear rise in hallucination as the monofact rate grows: the hallucination rate climbs from $\approx 0\%$ to $\approx 50\%$ when the monofact share increases from 0% to 100%. This trend roughly matches the factor-0.5 slope.

The middle right panel of Fig. 2 confirms that monofact rate and miscalibration naturally move in tandem without intervention. The right panel further shows that empirical KL divergence successfully tracks miscalibration, supporting its use as a practical proxy for our defined setting.

Fig. 3 reports mean bin-wise miscalibration for three monofact ranges *without any miscalibration injection*. As monofact rates and hallucination fall, miscalibration in the right-most (lowest-confidence) bin contracts towards zero and we observe a similar increase in *polarity*. What dampens hallucination organically is not simply the magnitude of miscalibration, but relative polarity of bin-wise miscalibration gained from sampling training data using Pareto distributions.

B. Miscalibration Injection as a Hallucination-Reduction Mechanism. With monofact rates fixed, Fig. 4 traces the effect of the upweighting algorithm for controlled miscalibration injection. Upweighting just 312 examples ($\approx 6\%$ of the corpus) could significantly increase overall miscalibration and KL divergence while halving hallucination in some cases (Fig. S4 in SI appendix shows additional results). Miscalibration and KL divergence return to its baseline when k nears $|S|$ due to normalization. The hallucination reduction effect is most pronounced in high-monofact situations. We find that our miscalibration injection method pushes bin-wise polarity even higher while increasing the total magnitude of miscalibration and KL divergence.

4. Supervised Fine-tuning Methodology

To test whether the theoretical relationship between monofact rates, miscalibration, and hallucination extends to modern language models, we fine-tuned Transformer models on synthetic biographical data with similarly controlled fact-frequency distributions. This approach allows us to systematically manipulate monofact rates while measuring hallucination in naturalistic text generation—a setting more representative of real-world applications than n-gram models.

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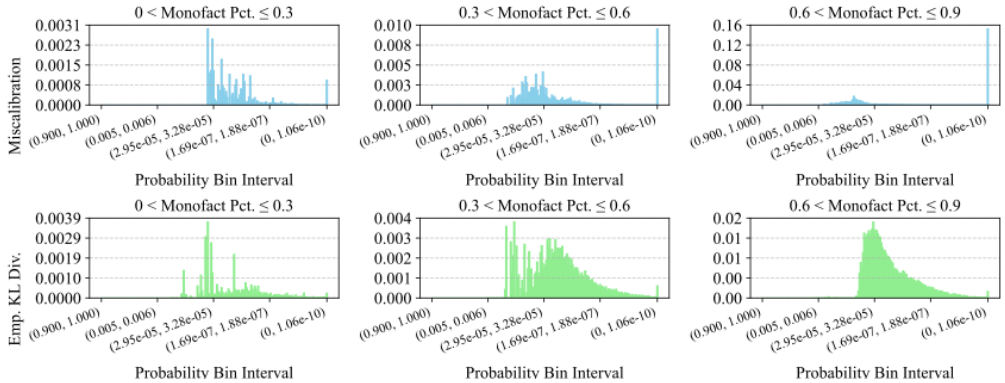

Fig. 3. Top: Average miscalibration per probability bin (binning created by a logarithmic binning strategy with ϵ = 0.1) for three monofact percent ranges. **Bottom:** Average empirical KL divergence per probability bin for three monofact percent ranges.

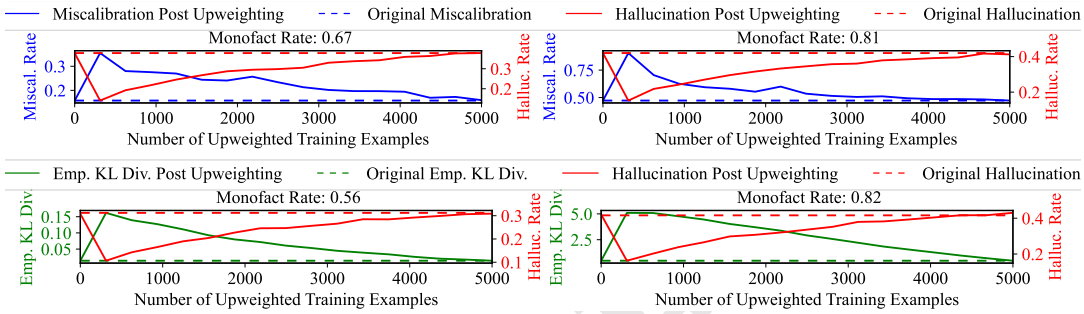

Fig. 4. Top: Relationship between miscalibration (blue line, left y-axis) and hallucination rates (red line, right y-axis) for select fixed monofact rates. Dotted lines indicate metrics prior to any interventions. Each subplot shows how miscalibration and hallucination evolve as we duplicate token occurrence for more and more statements from the training data (size of 5,000). **Initial miscalibration** (before upweighting intervention) is shown as the leftmost point of each blue curve at $k = 0$ upweighted examples. **Bottom:** Relationship between empirical KL Divergence (green line, left y-axis) and hallucination rates (red line, right y-axis) for select fixed monofact rates.

We generated 10,000 unique biographies, each containing seven attributes: name, birth date, hometown, college, major, job title, and employer. Each attribute was sampled from pools of varying sizes to create natural frequency imbalances—for instance, 100,000 unique names versus 174 unique college majors—mirroring the long-tailed distributions observed in real-world knowledge bases. Biography text was assembled using four randomly selected template variants (e.g., "[name] was born on [date]" versus "[name]'s birthday is [date]") to introduce linguistic diversity while preserving factual content. Data pipeline details could be found in SI Appendix.

Following recent work showing that power-law-distributed training data accelerates factual learning, we controlled monofact rates by sampling biographies according to Pareto distributions with varying shape parameters γ. Lower γ values produce datasets where most biographies appear multiple times (low monofact rate), while higher γ values approach uniform sampling where nearly every biography appears only once (high monofact rate). This generates training sets spanning monofact rates from 15.5% to 99.8% while maintaining a fixed vocabulary and total training budget.

All models are fine-tuned using standard cross-entropy (negative log-likelihood) loss on next-token prediction:

$$\mathcal{L} = -\frac{1}{|S|} \sum_{(x_1,\ldots,x_T)\in S} \sum_{t=1}^{T} \log p_\theta(x_t \mid x_{<t}),$$

where S is the training corpus and θ denotes model parameters. During upweighting, selected examples contribute to the loss multiple times per epoch according to their duplication factor.

A. Upweighted Training. We conduct SFT traing on the T5-SMALL, T5-LARGE (16), GPT2-MEDIUM, and GPT2-LARGE (17), tracking four metrics throughout training: hallucination rate (measuring whether generated biographies contain correct attributes), miscalibration (total variation distance between p and g distributions), empirical KL divergence, and forced-generation inaccuracy (testing factual recall through prompted completion). This experimental design allows us to measure how data frequency distributions affect both what models learn and how confidently they express that knowledge. Specifically:

- **Inaccuracy**: The error rate on forced-generation factual recall. Given a prompt (e.g., "[Name]'s birthday is _____"), we measure whether the model's most likely completion matches ground truth. This captures errors during *prompted recall*.
- **Hallucination rate**: $|G \cap F|/|G|$, where G is the set of generated statements and F is the set of falsehoods. This captures errors during *free-form generation*, where the model produces plausible but false statements unprompted.

These metrics are not complements: a model may achieve low inaccuracy on prompted recall while exhibiting high hallucination during open-ended generation, or vice versa.

To test miscalibration injection as a hallucination mitigation strategy, we trained models under three upweighting regimes. The baseline regime trains without any sample reweighting, allowing us to measure the natural relationship between monofact rates and hallucination. In the last-stage upweighting regime, we train normally for the first two-thirds of training, then upweight a small fraction (5%–30%) of training examples by duplicating them

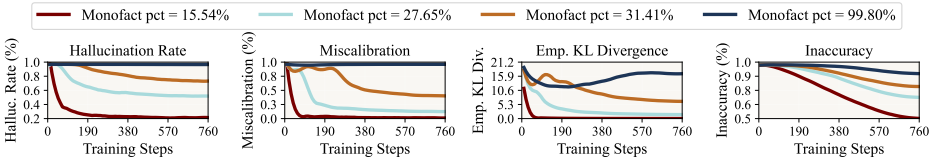

Fig. 5. (T5-Small) Four training statistics (from **Left** to **Right**: hallucination, miscalibration, KL divergence, inaccuracy) for every 5 steps during supervised fine-tuning for four training regimes with varying monofact rates but a fixed sample size of 10,000.

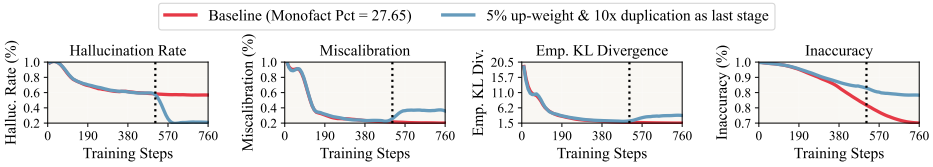

Fig. 6. (T5-Small) Four training statistics (from **Left** to **Right**): hallucination, miscalibration, KL divergence, inaccuracy) for every 5 steps during supervised fine-tuning by upweighting 5% of the training data 10 times as the *last* stage of fine-tuning for a training sample with 27.65% monofact rate. Dashed black line indicates when *last-stage* upweighting injection takes place. **Initial miscalibration** prior to intervention is visible as the value before the vertical dashed line.

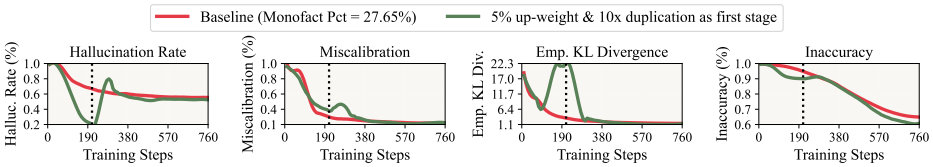

Fig. 7. (T5-Small) Four training statistics (from **Left** to **Right**): hallucination, miscalibration, KL divergence, inaccuracy) for every 5 steps during supervised fine-tuning by upweighting 5% of the training data 10 times as the *first* stage of fine-tuning for a training sample with 27.65% monofact rate. Dashed black line indicates when *first-stage* upweighting injection ends and normal SFT starts.

multiple times (5× to 10×) during the final training stage. This deliberately injects miscalibration after the model has learned most factual content. Conversely, the first-stage upweighting regime applies duplication early in training, then returns to standard training — testing whether miscalibration injection works only as a final intervention or throughout learning. By comparing hallucination rates across these three regimes while holding monofact rates constant, we isolate the causal effect of intentional miscalibration on model outputs.

Additional information on training configurations, template specifications, and evaluation protocols appear in SI Appendix.

5. Supervised Fine-tuning Results

A. Upweighted Training Results. In this section, we present detailed results for T5-SMALL and then discuss findings across additional model architectures and scales. Results are consistent across all four LLM models and model sizes unless otherwise noted.

Without any miscalibration injection, Fig. 5 illustrates for that low-monofact data facilitates faster convergence, hallucination prevention, and better overall accuracy, echoing the finding that data sampled from a Pareto distribution enables shorter knowledge acquisition plateaus (3).

Fig. 6 tracks SFT metrics when miscalibration is injected at *step 510* on a dataset with a 27.65% monofact rate. Miscalibration and empirical KL divergence rise and remain elevated, yet the hallucination rate drops by roughly 40%. Overall inaccuracy dips slightly after the intervention and then stabilizes at roughly pre-injection level, indicating that accuracy is not harmed by miscalibration injection. However, there is a clear hallucination-accuracy trade-off compared with the uninjected baseline: inaccuracy continues to inch downward while hallucination stays higher; the optimizer therefore favors overall accuracy improvements unless we intervene.

B. Model Architecture Discussion. To understand why last-stage upweighting reduces hallucination, we tested the same intervention at the start of training (Fig. 7) for Encoder-Decoder models T5-SMALL and T5-LARGE. Early-stage upweighting leaves terminal hallucination rates unchanged but reduces overall error slightly (1 - 3%). We hypothesize that early injection aids initial learning of generic data representations, while late-stage injection reinforces previously encountered examples, thereby suppressing hallucination more effectively in encoder-decoder architectures.

This pattern reverses for decoder-only models (Table. 1). Both GPT2-MEDIUM and GPT2-LARGE benefit more from first-stage upweighting for hallucination mitigation, leading to reduction of 8 - 10%. This architectural dependence likely reflects differences in how bidirectional versus autoregressive attention mechanisms incorporate distributional biases during learning. Practitioners should therefore select upweighting timing based on their model architecture, with last-stage interventions for encoder-decoder models and first-stage interventions for decoder-only models.

Combining first and last-stage upweighting does not preserve the benefits observed from either intervention alone. Across all architectures tested, combined upweighting at both stages yields no improvement in hallucination rates, suggesting that excessive upweighting could disrupt the model's learned representations.

C. Practical Guidance for Upweighting Interventions. A key question remains: how do we decide on the magnitude and size of the injections? Fig. S2 in SI Appendix surveys a few upweighting configurations, formed by crossing three sample shares (5,15,30%) with three duplication multipliers (1,5,10), under progressively higher monofact rates (15.5, 27.6, 31.4%).

To translate our empirical findings into actionable recommendations, we conducted two complementary analyses of upweighting

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